

Akshay Kolwalkar

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Education

University of Toronto (St. George)

Sep. 2024 - Apr. 2028

BASc, Mechanical Engineering + PEY/Co-op

Toronto, ON

Minors (pursuing): Robotics and Mechatronics; Engineering Business

Relevant coursework (completed/current year): Fluid Mechanics I, Thermodynamics, Heat & Mass Transfer, Kinematics & Dynamics of Machines, Mechanics of Solids I-II, Numerical Analysis & Computational Methods, Manufacturing Engineering

Technical Skills

Design & Analysis: SolidWorks (CSWA), MATLAB, Simulink, STK 12, SPICE/SpiceyPy; Machining Certificate

Programming & Data: Python, C++, TypeScript, PyTorch, NumPy, pandas, numerical methods, Docker, regression testing

Robotics & Controls: ROS 2, RViz, AutoDrive, Gym, NVIDIA Jetson NX, SAC reinforcement learning, attitude estimation

Work Experience

GE Appliances

Jun. 2025 - Aug. 2025

Manufacturing Production Intern

Louisville, KY

- Analyzed **12 months of plant performance statistics** and proposed assembly-process designs projected to increase **units produced by 6%**, using hands-on production experience to prioritize improvements in throughput and assembly quality.
- Developed production-quality design recommendations with a **projected 16% reduction in defect rate**, using historical plant data to assess improvement potential and present findings to area manufacturing engineers for technical review.
- Proposed equipment-maintenance improvements estimated to reduce **routine maintenance time by 12%**, combining historical plant-performance analysis with hands-on equipment operation and maintenance experience on the factory floor.

Louisville Automation and Robotics Research Institute (LARRI)

Student Researcher

Aug. 2025 - Present

- Improved **simulated lap times by 23%** over the previous year's autonomous model through autonomous-driving algorithm development with **Dr. Sabur Baidya**, supporting autonomous-racing research for the **2027 F1TENTH competition**.
- Reduced **simulated collision rate by 85%** while developing **Soft Actor-Critic (SAC) policies in PyTorch and Gym**, evaluating autonomous-driving behavior in **AutoDrive 3D simulation** to improve racing pace and collision avoidance.
- Supported an **NVIDIA Jetson NX** race-car research platform with **ROS 2, RViz and Docker**, using simulation, vehicle state visualization and software environment setup to support autonomous-control development and policy evaluation.

Engineering Teams and Projects

University of Toronto Aerospace Team - Space Systems

May 2025 - Present

Attitude Determination and Control Systems

- Developed a **MATLAB/Simulink CubeSat ADCS simulator** integrating orbit dynamics, sensors, attitude estimation and control, with **4 reaction wheels and 3 magnetorquers**, quaternion kinematics and modeled actuator limits.
- Executed a **6,000-second mission with 600,001 states**; corrected star-tracker noise orientation and checked quaternion normalization, angular-momentum conservation and timestep stability through regression diagnostics.
- Built **TypeScript/Three.js telemetry replay** preserving epochs, mode events and delivered actuator torque; passed **15 unit tests and 10 desktop/mobile interaction tests** using exported MATLAB simulation data.

NASA Space Apps Hackathon - NEOScope

Oct. 2025

Awards: Local Impact and Most Inspirational

Toronto, ON

- Delivered **NEOScope in 2 days** at NASA Space Apps Toronto, earning **Local Impact and Most Inspirational** awards at an event with **140 in-person participants** through an interactive asteroid-impact simulation.
- Processed **40,000+ near-Earth object records** with **NumPy and pandas** and modeled **40 asteroid orbits**, applying Keplerian mechanics and atmospheric-entry calculations to assess trajectories, impact effects and mitigation.
- Integrated a **React/Three.js frontend** with a **Python/Flask backend** and NASA small-body and Sentry APIs, connecting orbital data to **3D orbit visualization and 2D impact simulations** for exploring mitigation strategies.

University of Toronto Formula Racing (UTFR)

Dec. 2025 - Present

Aerodynamics & Vehicle Dynamics Team Member

- Contribute to UT27 aerodynamics discussions spanning **3 core components: front wing, undertray and rear wing**, using team **CATIA and STAR-CCM+** studies to assess downforce, drag and aerodynamic balance.
- Support technical review of **mesh-independence and flap-angle studies** across **2 flow conditions: straight-line and cornering**, informing aero-package refinement and simulation reliability.
- Contribute to the UT27 **driver-in-the-loop (DiL) simulator** on **VI-Grade DriveSim and rFpro**, integrating Simulink powertrain models and **UM14+ tyres**, and correcting steering-torque and tyre-fit terms (Mz vs camber, QCZ1) flagged in driver feedback and FRF/slalom comparisons.